

Customer Churn Analysis Project Report

Introduction

This project focuses on analyzing customer churn behavior using data analysis and visualization techniques to identify key factors affecting customer retention and business performance.

Abstract

The project combines Python (Jupyter Notebook) for data preprocessing and exploratory data analysis with Power BI for interactive dashboard visualization of churn patterns.

Tools Used

Python (Pandas, NumPy, Matplotlib, Seaborn), Jupyter Notebook, Power BI, Excel Dataset.

Steps Involved in Building the Project

1. Imported and explored dataset in Jupyter Notebook. 2. Performed data cleaning and handled missing values. 3. Conducted exploratory data analysis (EDA) to identify churn patterns. 4. Exported cleaned data to Power BI. 5. Created KPIs and dashboards including churn rate, active customers, and churn drivers. 6. Added insights and finalized report.

Conclusion

The project identifies key churn factors such as contract type, monthly charges, and internet service. The combination of Python analysis and Power BI visualization helps in better understanding customer behavior and improving retention strategies.